LAB # 11
To Explain and Reproduce the Working of Stepper and DC motor on STM32F407 Trainer Board using C Programming.

Objectives

- To explain the working of Stepper and DC motor with the STM32F407 using trainer board.
- To modify the program and display its output on STM32F407 trainer board using C programming.
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Pre-Lab Exercise

Read the details given below in order to comprehend the basic operation of buzzer and interfacing of buzzer with microcontroller.

Introduction to Motors

Motors (Stepper and DC) convert electric energy to mechanical motion. The type of motor chosen for an application depends on the characteristics needed in that application. These include:

- How fast you want the object to move?
- The weight, size of the object to be moved.
- The cost and size of the motor.
- The accuracy of position or speed control needed

Stepper motor

It offers slow, precise rotation and easy control. It has advantage over servo motors or DC motors in positional control where servos require a complicated feedback mechanism. A stepper motor has positional control via its nature of rotation by fractional increments. Suited for 3D printers and similar devices where position is fundamental.

Stepper has many electromagnets. The rotor carries a set of permanent magnets, and the stator has the coils. Stepper controlled by sequential turning on and off of electromagnet coils. Each pulse moves another step, providing a step angle. It can be moved to and held in a desired position. It can be rotated continuously at a controlled speed.

- For 4 steps per revolution
  - 360/4 = 90o step angle
- If motor of 100 steps per revolution, then
  - 360/100 = 3.6 o step angle
  - Because Rotor not only has Only 2-sided Magnet, it has multiple teeth
  - After 4 steps, rotor moves only one tooth pitch
- In Stepper Motor with 100 steps per revolution, rotor would have 25 teeth
  - 4 x 25 teeth = 100 steps needed to complete one revolution

A stepper drive is the driver circuit that controls how the stepper motor operates. Stepper drives work by sending current through various phases in pulses to the stepper motor. There are four types: wave drive, full step drive, half step drive and microstepping drive.

Wave drive works with only one phase turned on at a time. Consider the illustration in Figure 11.1. When the drive energizes pole A (a south pole) shown in green, it attracts the north pole of the rotor. Then when the drive energizes B and switches A off, the rotor rotates 90° and this continues as the drive energizes each pole one at a time.

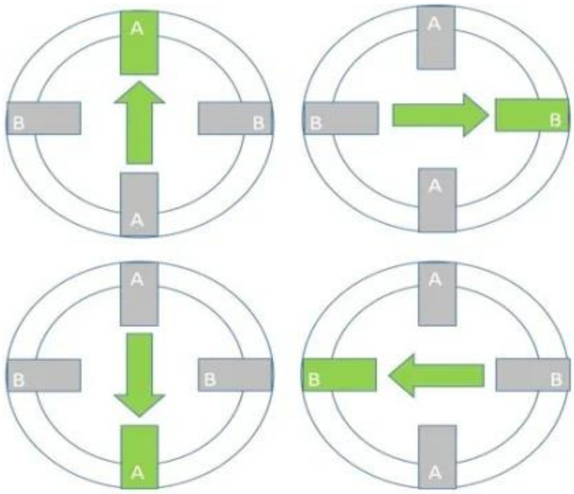

Figure 11-1: Wave drive

Engineers rarely use wave driving it is inefficient and provides little torque, because only one phase of the motor engages at a time.

Full step drive has its name because two phases are on at a time. As seen in Figure 11.2, if the drive energizes both A and B poles as south poles (shown in green), then the rotor's north pole attracts to both equally and aligns in the middle of the two. As the energizing sequence continues on like this, the rotor continuously ends up aligning in-between two poles.

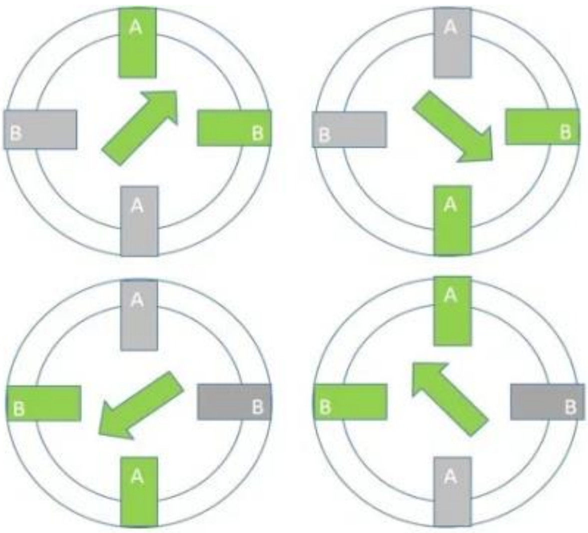

Figure 11-2: Full step drive

Two-phase-on driving gets no finer resolution than one-phase on, but it does produce more torque.

Half step drive has its name for the way the drive energizes either 1 or 2 phases at any specific time. In this driving method, also known as half-stepping, the drive energizes pole A (shown in green) … then energizes poles A and B … then energizes pole B … and so forth. This can be seen in Figure 11.3

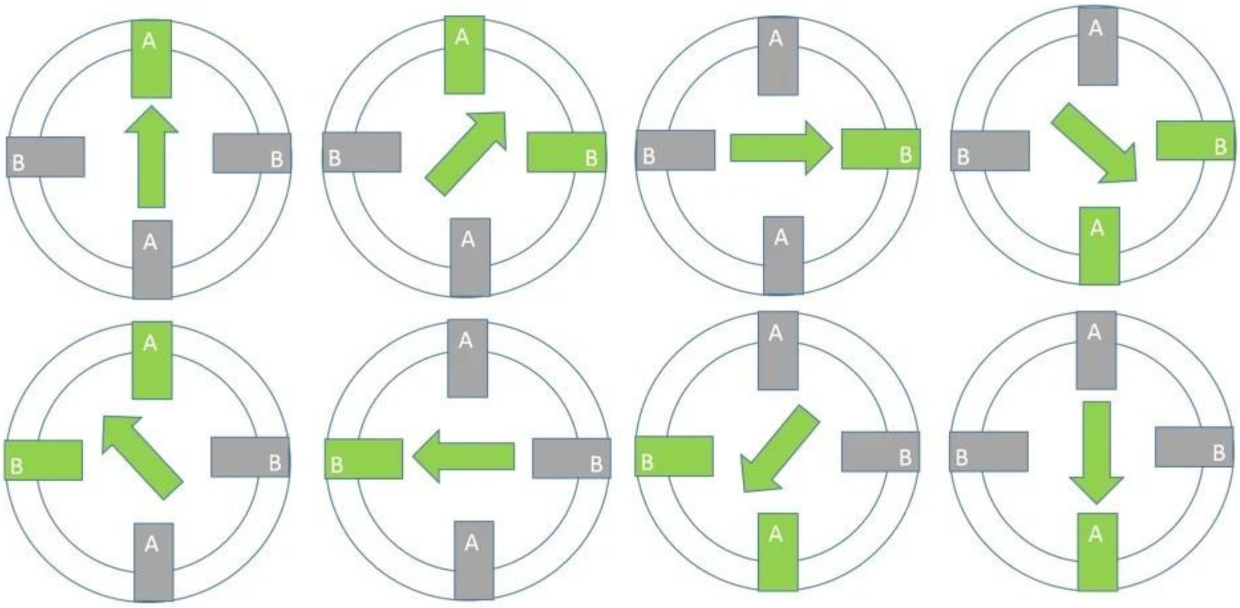

Figure 11-3: Half step drive

DC motor

The electric motor operated by dc is called dc motor. It is the most common and cheapest motor available for use. This is a device that converts DC electrical energy into a mechanical energy. Figure 11.4 shows the physical motor and the conversion phenomena. The important features of a DC motor are as follows:

- Powered with two wires from source
- Draws large amounts of current
- Cannot be wired straight from Controller
- Does not offer accuracy or speed control
- Fast, continuous rotation motors
- Used for anything that needs to spin at a high RPM e.g., car wheels, fans etc.

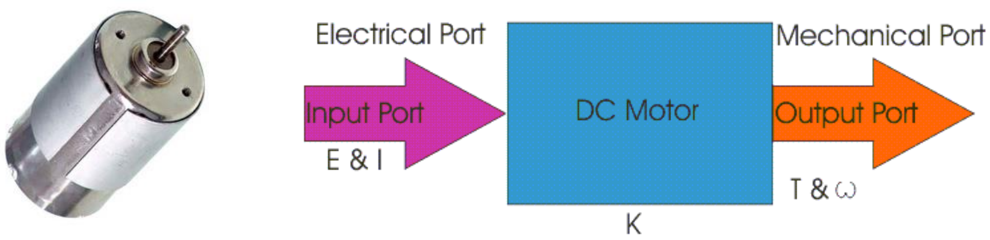

Figure 11-4: DC Motor

The interfacing of DC motor with microcontroller requires a bridge circuitry in between called as an H-Bridge. It makes use of the switching transistors and allows the bi-directional rotation of the DC motor through switching contacts. Figure 11.5 shows the internal circuitry of the H-bridge constructed with MOSFET switches and is further demonstrated with two directional current flow in Figure. 11.6 with switching scheme in Table 11-1.

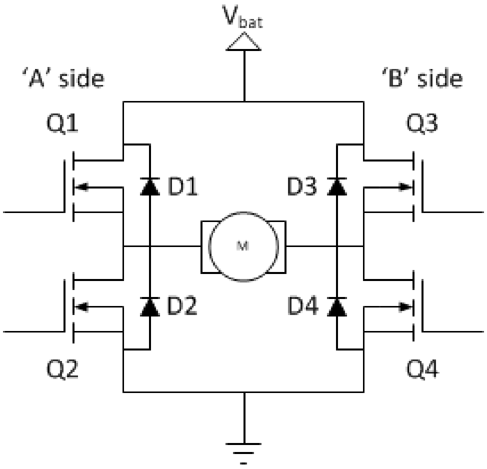

Figure 11-5: H-Bridge internal circuitry

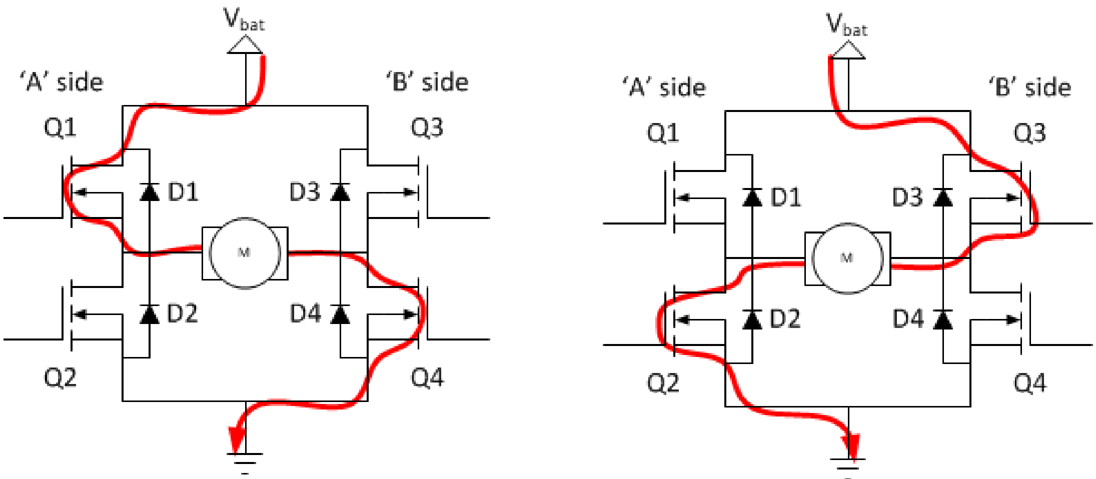

Figure 11-6: H-Bridge directional flow

Table 11-1: H-Bridge mode

Q1	Q2	Q3	Q4	mode
close	open	open	open	motor coasts
close	open	open	close	forward motoring / move right
close	open	close	open	motor brakes (zero potential voltage)
open	close	open	open	motor coasts
open	close	open	close	motor brakes (zero potential voltage)
open	close	close	open	reverse motoring/ move left
open	open	open	open	motor coasts
open	open	open	close	motor coasts
open	open	close	open	motor coasts

In-Lab Exercise

Task-1 Write a complete C program to interface a stepper motor with STM32F4

Write a C-program to drive a stepper motor using HAL libraries. The stepper motor is interfaced with Port B's bit 0.1,2,3.

Task-2

Two push buttons are connected with STM32, which are used to rotate a DC motor. Design a scenario which rotates the DC motor according to the following criteria

Push Button 1	Push Button 2	DC Motor Status
LOW	LOW	STAND STILL
LOW	HIGH	CLOCKWISE
HIGH	LOW	ANTI-CLOCKWISE
HIGH	HIGH	PULSE (2 sec) STOP (2 sec) REPEAT

Rubric for Lab Assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed all assigned tasks independently, strictly followed all health and safety protocols, and presented the results clearly and accurately.	4	
Good	The student completed the assigned tasks with minimal guidance, mostly followed health and safety protocols, and presented the results appropriately.	3	
Average	The student partially completed the assigned tasks, inconsistently followed health and safety protocols, and presented partial or unclear results.	2	
Worst	The student did not complete the assigned tasks, frequently disregarded health and safety protocols, and failed to present results.	1	

Instructor Signature: ____________________________ Date: _________________